Recommendation Problems and Classical Methods

CP4285 Modern Recommendation Systems — Week 01

Scan for the slides
soc-n.us/cp4285-t2610-w01

CP4285 Instruction Team

11 Aug 2026

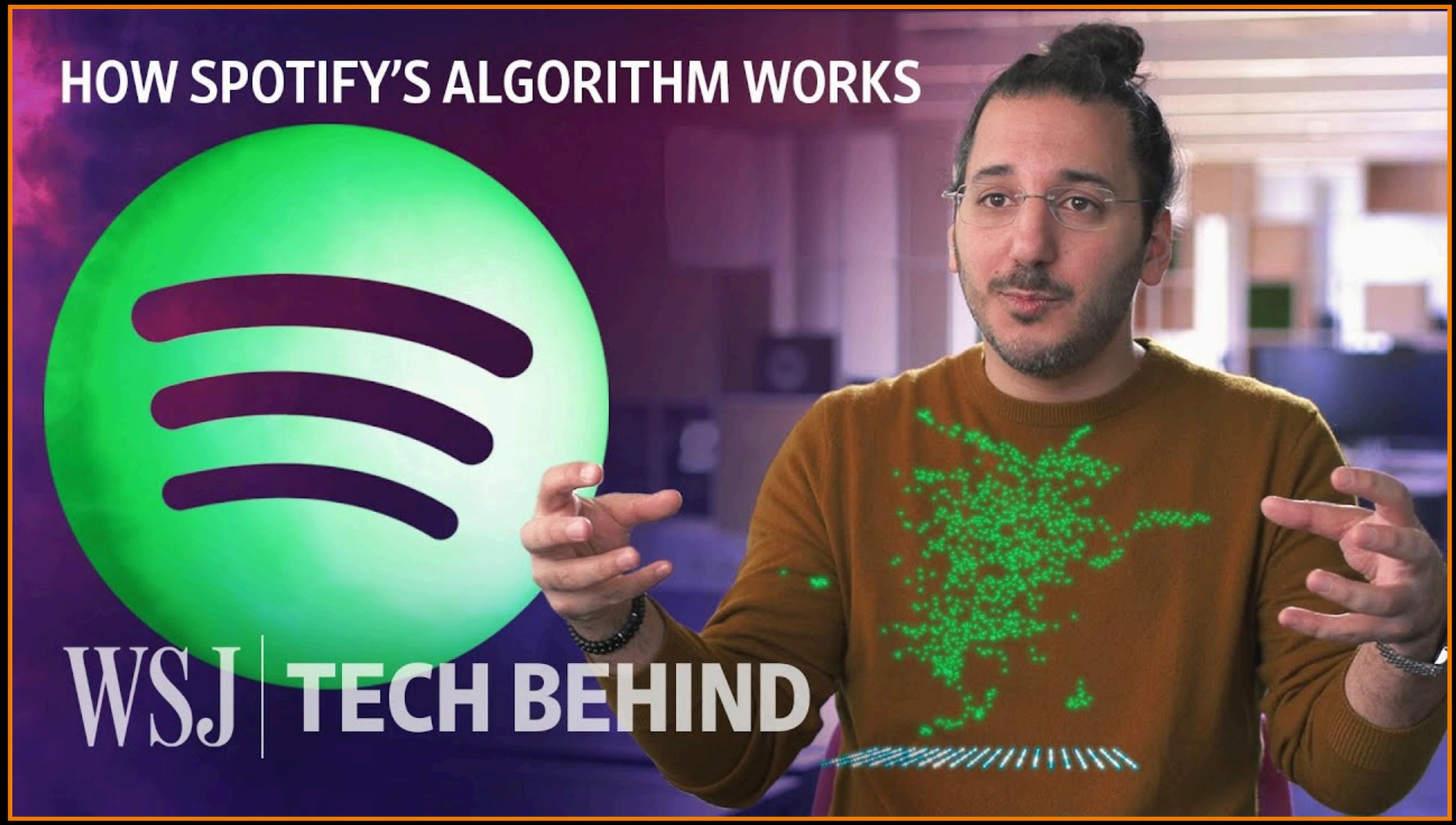
01 Overview

CHANGELOG

- **10 Aug 2026** — Default assignment deadlines are Mon, 23:59 SGT.
- **10 Aug 2026** — Project teams are 2–4 students; Min may adjust teams to support diversity.
- **10 Aug 2026** — Added essay and Project Mini-team Declaration deadlines.



Video: Spotify Recommendations



Recommendations Everywhere

Everyday surfaces

Video, music, shopping, news, jobs, dating, maps, social feeds, and course resources.

Inputs — behaviour, context, item attributes, social signals
Outputs — a shortlist, ranking, or exposure decision

> ⚠ **Warning**
>
> A recommender does not merely retrieve information. It helps decide what receives attention — and what remains unseen.



CP Pilot Course

CP4285 is a **pilot** in the **CP428x Emerging Topics in Computer Science** series.

- We test and improve content, assessment, labs, tutorials, and delivery for future cohorts.
- Students help by self-learning, building **labs and tutorials**, **slide notes** for more comprehensive study.
- And documenting processes, gaps, and improvements.

> ⚠ **Warning**
>
> This means that your class notes will rarely be ready before class. We have to create the materials from scratch.



Bandit Time: Objectives

Same data, different objective

Two systems use the same user data but optimise different objectives. What should we expect?

A. They should agree if they observe the same users and items.

B. They may rank different items because the objective changes what gets exposed.

C. They can differ only when one system has more user history.

D. Different objectives change scores, but not the ordering of items.

00:20

▶ Start Timer

Choose an answer; justify it in one sentence.

AI Voice Mode

We are using AI Voice Mode experimentally to investigate the **ethics of recommenders**: how an interactive system explains trade-offs, frames uncertainty, and responds when we question who receives attention.

It is a conversation partner for inquiry, not an authority or a production recommender. Listen for what it assumes, what it leaves out, and how its wording could shape our judgement.

👥 Find the Recommender

Audit one recommendation

1. Name one recommendation you received recently.
2. What signals might the system have used?
3. What objective might it be optimising?
4. Who or what might be left unseen?

Compare with a partner; share one observation.

03:00

▶ Start Timer

Ethics Thread

Recommendation systems allocate **attention, visibility, and opportunity**.

Historical data and engagement objectives can reproduce or amplify **bias**.

Personalisation raises questions about **privacy, transparency, exposure, fairness, and stakeholder impact**.

02 Schedule

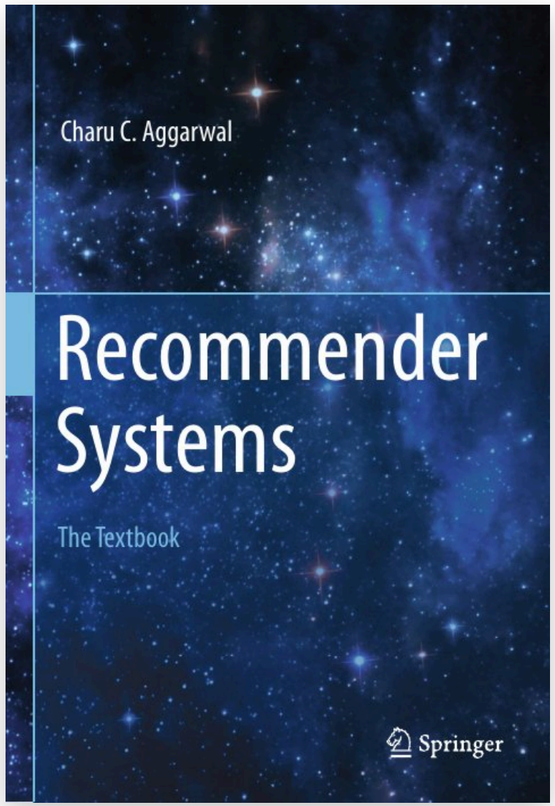


 Roadmap: Course Learning Outcomes

By the end of CP4285, you should be able to:

- **CLO 1:** Compare classical, content-based, collaborative-filtering, and hybrid methods.
- **CLO 2:** Implement matrix-factorization and neural recommenders.
- **CLO 3:** Design rigorous offline evaluation using ranking metrics.
- **CLO 4:** Analyse cold start and propose appropriate remedies.
- **CLO 5:** Critique systems through fairness, privacy, transparency, and stakeholder impact.
- **CLO 6:** Explain sequential, graph-based, multi-objective, and LLM-enhanced systems.
- **CLO 7:** Design and justify an end-to-end recommender pipeline.
- **CLO 8:** Communicate and defend technical designs and results.

Course Library

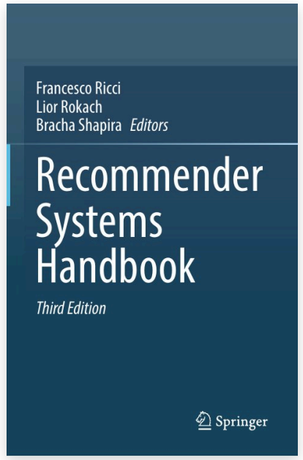


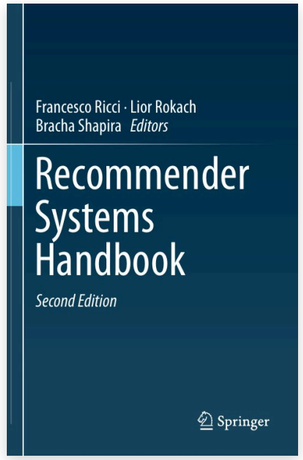
Core textbook

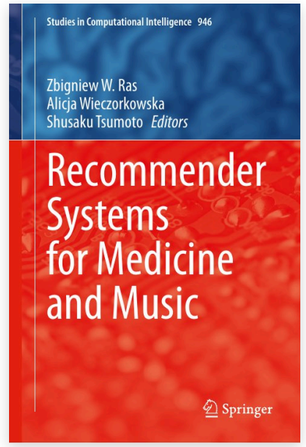
Charu C. Aggarwal, *Recommender Systems: The Textbook*

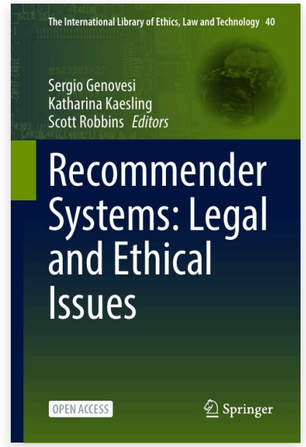
Focus of the first third of the course.

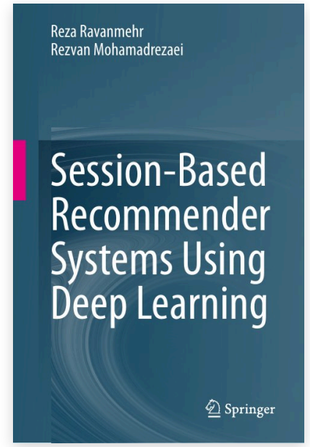
A course library for going deeper

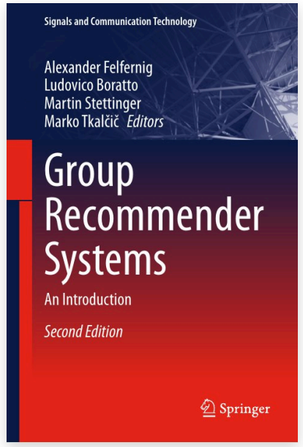


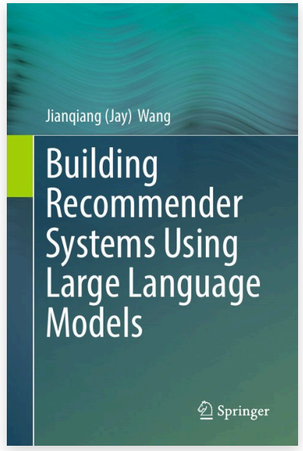


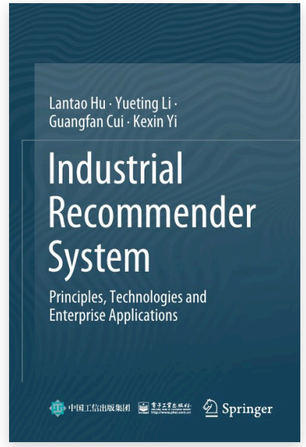


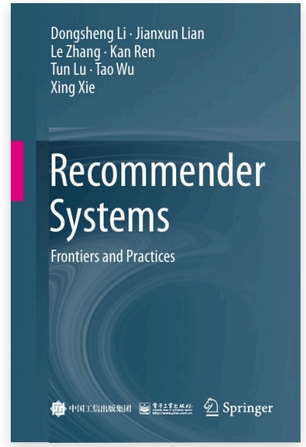


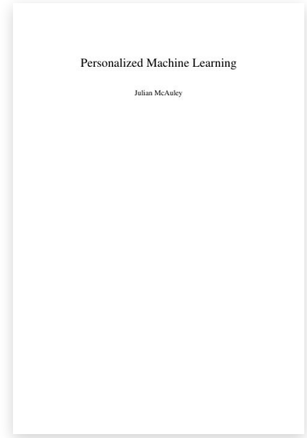












Weekly Topics

Week	Date	Topic
01	11 Aug	Recommendation Problems and Classical Methods
02	18 Aug	Latent Factor Models
03	25 Aug	Evaluation of Recommendation Systems
04	1 Sep	Neural Recommendation Models
05	8 Sep	Sequential and Session-Based Recommendation
06	15 Sep	Retrieval and Ranking Architectures



Weekly Topics (cont.)

Week	Date	Topic
—	22 Sep	*Recess Week — No Class*
07	29 Sep	Project Design Critique Workshop
08	6 Oct	Learning-to-Rank
09	13 Oct	Graph-Based Recommendation
10	20 Oct	Multi-Objective Recommendation
11	27 Oct	Exploration and Online Learning
12	3 Nov	LLMs, Generative Recommendation, Research



Class Meetings

- Time: Every **Tue, 10:00–12:00 SGT**
- Venue: **Seminar Room 12, COM3-01-21**

Lectures will be recorded where technology permits — recordings are for **revision only**, not a substitute for attendance

There are **no tutorial sessions** for this course — We'll be building the materials for the next cohort.



03 Assignments & Grading



Grading Overview

Component	Weight
Class Participation	10%
Essays	20%
Group Project	30%
In-Class Quizzes	10%
Final Exam (Mon, 23 Nov 2026, 13:00–15:00)	30%

Class Participation (10%)

1. **In-lecture activities**
2. **Pre-lecture activities**
3. **Pre-Flight Survey** and **Midterm Survey**
   - *Pre-Flight Survey* (due Week 02; Mon, 17 Aug 2026 23:59 SGT)
   - *Midterm Survey* (due Week 08; Mon, 5 Oct 2026 23:59 SGT)

   Complete these on time — they are **free marks**

> ⚠ **Warning**
>
> Missing the Pre-Flight or Midterm Survey will directly reduce your participation grade. Don’t forget to do them!



Essays (20%)

Two individual take-home essays, each worth **10%**

- **Essay 1:** Due Week 03; Mon, 24 Aug 2026 23:59 SGT
- **Essay 2:** Due Week 08; Mon 5 Oct 2026 23:59 SGT

Short written analyses of recommender-system methods, evaluation choices, and social implications (8%). Randomised peer review counts towards the remaining 2%.

AI tools are **permitted as a resource** — but an AI declaration is **mandatory** for each submission

> **Note**
>
> Essays assess your own critical thinking. AI may assist but must not substitute your analysis. So do them on your own, and don’t let AI do the hard work of understanding; the process is more important than the outcome.



Project (30%)

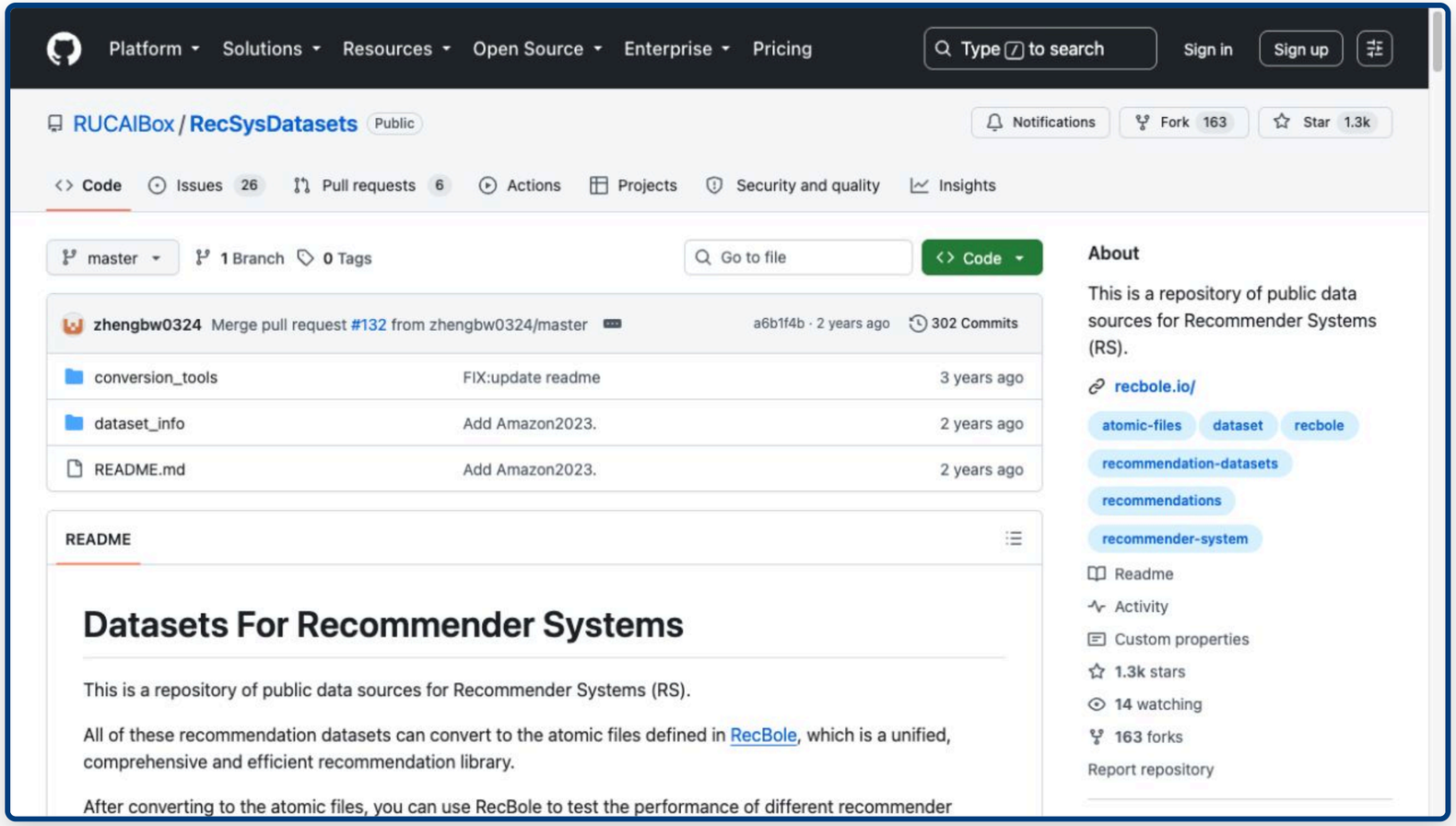
- Groups have **2–4 students**
- Define a clear problem anchored on a selected **recommendation system dataset**

Milestones:

- **Week 03:** Project Mini-team Declaration — Mon, 24 Aug 2026 23:59 SGT
  - Submit a self-formed team or indicate that you need instructor placement
  - Min has the final decision on group formation (may override student-formed groups)
- **Week 07:** Project Design Critique Workshop
- **Week 13:** Final Project Presentation at 29th STePS: Wed, 11 Nov
  - Teams presenting at 29th STePS do **not** submit a project report; other teams submit a report instead, due same Wed, 11 Nov 2026 23:59 SGT



Recommendation Dataset Catalogue



Click the screenshot to open the live dataset catalogue.

RUCAIBox. (n.d.). *RecSysDatasets* [GitHub repository]. github.com/RUCAIBox/RecSysDatasets

Quizzes (10%)

- Quizzes are **jointly created by peers and instructors**
- Peers contribute candidate questions, answers, and plausible distractors from their self-learning
  - Min will assign peers randomly to a week and section responsible.
  - Min will curate, verify, and finalise questions so that the assessments remains fair, accurate, and aligned with the course learning outcomes
- Combined worth **10%** of total marks
- **AI tools must not be used during in-class quizzes.**

> **Note**
>
> Co-creation makes quizzes part of the pilot course: students help build better learning materials for this cohort and future cohorts.



Authority Chain

Our slides will be the most authoritative source of truth.

While the we will try to get the slides synced with the website or Canvas, use information introduced in lecture as **the definitive source**.

There will likely be errors in the lecture, so check any doubts or questions with Min.

Unless a slide or Canvas announcement specifies otherwise, assignment deadlines are **Mondays, 23:59 SGT**.

> **Note**
>
> The slide deck built in Quarto and rendered through `reveal.js`. The website is also built Hugo and served using Github Pages. This means any student, member of the public, or AI can inspect all details of the course. Use this to your advantage.



☕ BREAK · THE RECOMMENDATION DESK



Mrs. Arabella Murray, librarian, works at her desk while five girls read at a nearby table.

Digital Public Library of America. (n.d.). *Photograph of a librarian at the Dunbar Branch Library, Athens, Georgia, 1944–1948* [Photograph]. Source

05:00

▶ Start Timer

04 Recommendation Problems and Decision Goals

- User–item matrices, feedback, and missingness
- Prediction, top-*k* ranking, and useful lists
- Signals and classical recommendation strategies
- Cold start and heavy-head/long-tail exposure
- Treatment lenses and safeguards



The User–Item Matrix

R	i_1	i_2	i_3	i_4	i_5
u_1	5	?	2	4	?
u_2	?	4	?	1	3
u_3	?	?	?	?	?
u_4	3	?	4	?	2
u_5	?	2	?	5	?
u_6	5	4	5	2	4
u_7	5	1	5	1	5
u_8	?	?	4	?	?

$$R \in \mathbb{R}^{|U| \times |I|}$$

- $u \in U$: user; $i \in I$: item
- r_{ui}: observed feedback
- Ω: observed (u, i) pairs

> ⚠ **Warning**
>
> A ? is usually **unobserved**, not a negative rating.

> 🤔 **To think about**
>
> What user types might be represented in the matrix?
> What other forms can “ratings” take?

From a Score to a Useful List

Prediction

Estimate a value such as $\hat{r}_{ui}$ for one user–item pair.

Question: “Will this user rate this film highly?”

Top-*k* ranking

Order candidate items and present a short list.

Question: “Which five films should appear first, now?”

Accurate scores do not automatically make a useful list.

> 🤔 **To think about**
>
> Does top-*k* apply to users as well as items?

Signals and Usefulness

Interaction signals
ratings · clicks · saves · purchases

Item signals
metadata · text · images

Context
time · device · place

Social and operational signals
friends · availability · constraints

> **Note**
>
> The right evaluation follows the intended usefulness — not just the available data.

A useful recommendation can be …

- **relevant** to the task
- **novel** or serendipitous
- **diverse** across the list
- **well covered** across a catalogue
- **explainable** and constraint-respecting



👥 Signal Hunt

Interaction signals
ratings · clicks · saves · purchases

Item signals
metadata · text · images

Context
time · device · place

Social and operational signals
friends · availability · constraints

What else could a recommender capture?

1. Name **three** additional signals.
2. Place each in a box at left — or argue for a new box.
3. For one signal, name a benefit and a privacy, fairness, or manipulation risk.

02:00

▶ Start Timer

Pairs: prepare one signal to report.
📶 to critique ethical standpoint.

👥 Complete the Classical Methods Table

In pairs, complete the four orange cells. Be ready to justify one answer from the evidence each method uses.

Approach	Main evidence	Useful when	Early limitation
Popularity	aggregate interactions	_____	reinforces the head
Content-based	item attributes + one user	features are meaningful	_____
Collaborative filtering	patterns across users	_____	sparse or cold data
Knowledge-based	requirements and constraints	history is weak	_____



Four Classical Starting Points

Approach	Main evidence	Useful when	Early limitation
Popularity	aggregate interactions	little personal data	reinforces the head
Content-based	item attributes + one user	features are meaningful	narrow, familiar lists
Collaborative filtering	patterns across users	history is available	sparse or cold data
Knowledge-based	requirements and constraints	history is weak	needs elicitation



When the Matrix Has No Answer

❄️ Cold start

- **New user:** no observed preferences
- **New item:** no interaction evidence
- **New system:** few observations overall

A response is a design choice

- ask for requirements or examples
- use item metadata
- use a transparent popularity fallback
- reserve exposure for new or long-tail items



Bandit Time: What should this system optimise?

A campus library has sparse borrowing history and a student asks for “a short, accessible book on climate migration.”

A. Maximise predicted rating from the sparse matrix only.

B. Use stated requirements and explain the match.

C. Show the globally most-borrowed books.

D. Remove every item with little interaction history.

00:20

▶ Start Timer

“What types of users are there?”





Key Users to Consider

We'll use **card suits** as *User Archetypes* (treatment lenses) to reason about the outcomes. These are not demographic labels or permanent personalities. They are **situational lenses** for asking what a recommender should protect or provide.

User Archetype	Ask the system to consider
♣ **Clubs — adversarial / shill**	manipulation, coordinated abuse, robustness
♥ **Hearts — social**	social evidence, influence, conformity risk
♦ **Diamonds — novelty seeker**	freshness, diversity, serendipity, new-item exposure
♠ **Spades — loyalist / metadata-driven**	stable attributes, filters, consistency, explanation





Four Different Safeguards

♣ Clubs

Can coordinated ratings force a target item into the list?

♦ Diamonds

Does the list include genuinely new, long-tail, or emerging options?

♥ Hearts

Does a social cue inform the user without creating unwanted exposure or conformity?

♠ Spades

Can the user keep an important attribute fixed and understand why an item appears?

“Good recommendation” changes when the risk, task, and user situation change.



👥 Reprise: Find the Recommender

Return to the recommendation you audited in Section 01.

1. Name the likely signal and objective.
2. Is it predicting a response or ranking a shortlist?
3. Choose one user archetype to critique: a requirement, diversity/coverage target, explanation, or abuse check.

03:00

▶ Start Timer

Pairs: write a four-line audit.
📶 **critique one omitted stakeholder or assumption; challenge its answer.**

05 Neighborhood-Based Collaborative Filtering (CF)

- Rating types, sparse matrices, and long-tail evidence
- User–user and item–item neighbourhoods
- Similarity measures: Pearson, raw cosine, adjusted cosine, and confidence
- Worked examples: from shared evidence to a similarity score
- From a chosen neighbourhood to a prediction



Rating Types

- Continuous ratings
- Forced-choice ratings
- Ordinal ratings
- Binary ratings
- Unary ratings
- Explicit ratings
- Implicit ratings

> ⓘ **Note**
>
> **Missing-value imputation:** do not treat an unobserved explicit rating as a default score; pre-substitution can introduce bias.

> 🤔 **To think about**
>
> How do missing values impact training and testing in recommendation systems?

The Long Tail

- A small heavy head receives most ratings.
- The long tail holds many items with little evidence.
- Thin overlap makes neighbourhood similarity uncertain.
- Head-only ranking can reinforce what is already visible.

The tail is under-observed, not empty.

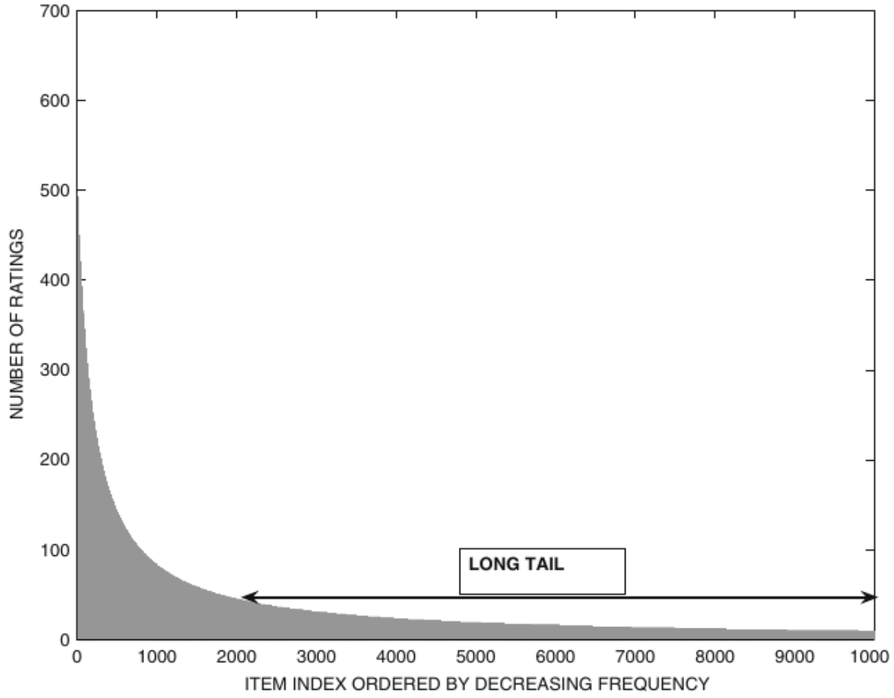


Figure 2.1: The long tail of rating frequencies

Source: Aggarwal (2016), p. 33.

From a Sparse Matrix to a Neighborhood

User	i_1	i_2	i_3	i_4	i_5
u_1 (u)	5	?	2	4	?
u_6 (v)	5	4	5	2	4

Mutual evidence

$$I_u \cap I_v = \{i_1, i_3, i_4\}$$

User-based CF

Find users who rate overlapping items similarly.

Item-based CF

Transpose the question: find items that receive similar patterns of ratings.

Standard notation

- I_u: items observed for user u
- U_i: users who observed item i
- $I_u \cap I_v$: shared evidence for a user pair
- $P_u(j)$: the neighbours of u who rated target item j

> ⚠ **Similarity is not a fact**
>
> It is an estimate from a partial overlap. With sparse data, two users may have no common ratings at all.



Similarity I: Pearson Correlation

$$\text{Pearson}(u, v) = \frac{\sum_{k \in I_u \cap I_v} \left(r_{uk} - \mu_u\right)\left(r_{vk} - \mu_v\right)}{\sqrt{\sum_{k \in I_u \cap I_v} \left(r_{uk} - \mu_u\right)^2} \sqrt{\sum_{k \in I_u \cap I_v} \left(r_{vk} - \mu_v\right)^2}}$$

Overlap: only mutually rated items contribute.

Centred Rating: rating minus that user's mean, so generous and harsh raters are put on a comparable baseline.

Normalisation: turns aligned variation into a bounded correlation-like score.

Use it when

Rating-scale differences matter.

Watch for

Very small overlap can create an unstable correlation.



Worked Example I: Pearson Correlation

User–user example: users 1 and 6

Excerpt from the user-item matrix R

User	i_1	i_2	i_3	i_4	i_5
u_1	5	?	2	4	?
u_6	5	4	5	2	4

Step 1 — begin with the same overlap

$$I_{u_1} \cap I_{u_6} = \{i_1, i_3, i_4\}$$

Step 2 — subtract each user's own mean

$$\mu_1 = \frac{11}{3}, \quad \mu_6 = 4$$

$$\mathbf{s}_1 = \left(\frac{4}{3}, -\frac{5}{3}, \frac{1}{3}\right), \quad \mathbf{s}_6 = (1, 1, -2)$$

Step 3 — correlate the centred vectors

$$\text{Pearson}(u_1, u_6) = \frac{-1}{\sqrt{14/3}\sqrt{6}} = -0.189$$

Similarity II: Raw Cosine

$$\text{raw-cosine}(u, v) = \frac{\sum_{k \in I_u \cap I_v} r_{uk}\, r_{vk}}{\sqrt{\sum_{k \in I_u \cap I_v} r_{uk}^2}\ \sqrt{\sum_{k \in I_u \cap I_v} r_{vk}^2}}$$

Overlap: compare only entries both users supplied.

Dot Product: large ratings in the same positions increase similarity.

Vector Length: corrects for the overall magnitude of each observed rating vector.

Use it when

Raw magnitude is meaningful or ratings are already on a comparable scale.

Watch for

It does not remove a user's tendency to rate high or low.



Worked Example II: Raw Cosine

Users 1 and 6

Excerpt from the user-item matrix R

User	i_1	i_2	i_3	i_4	i_5
u_1	5	?	2	4	?
u_6	5	4	5	2	4

> 🤔 **To think about**
>
> What are the differences between cosine and Pearson here?

Step 1 — keep only shared ratings

$$I_{u_1} \cap I_{u_6} = \{i_1, i_3, i_4\}$$

Step 2 — form the two rating vectors

$$\mathbf{r}_1 = (5, 2, 4) \qquad \mathbf{r}_6 = (5, 5, 2)$$

Step 3 — normalise their dot product

$$\text{raw-cosine}(u_1, u_6) = \frac{43}{\sqrt{45}\sqrt{54}} = 0.872$$

Similarity III: Adjusted Cosine for Items

$$\text{adjusted-cosine}(i,j) = \frac{\sum_{u \in U_i \cap U_j} s_{ui}\, s_{uj}}{\sqrt{\sum_{u \in U_i \cap U_j} s_{ui}^2}\ \sqrt{\sum_{u \in U_i \cap U_j} s_{uj}^2}} \qquad s_{ui} = r_{ui} - \mu_u$$

Shared Raters: users who rated both items.

User-Centred Rating s_{ui}: remove each rater's personal baseline before comparing item columns.

Normalisation: compare the centred item vectors fairly.

Use it when

Building item-to-item neighbourhoods from explicit ratings.

Watch for

It still needs sufficient shared raters; centring does not solve sparse overlap.

🤔 **To think about**

What makes this an "adjusted" metric? Why don't we need a similar adjustment for user-based cosine?

Worked Example III: Item–Item

Items i_1 and i_5 after user mean-centring

Original user-item matrix R

R	i_1	i_2	i_3	i_4	i_5
u_1	5	?	2	4	?
u_2	?	4	?	1	3
u_3	?	?	?	?	?
u_4	3	?	4	?	2
u_5	?	2	?	5	?
u_6	5	4	5	2	4
u_7	5	1	5	1	5
u_8	?	?	4	?	?

User	$\mathbf{s}_{i_1}$	$\mathbf{s}_{i_5}$
4	0	−1
6	1	0
7	1.6	1.6

$$\text{adjusted-cosine}(\mathbf{s}_{i_1}, \mathbf{s}_{i_5}) = \frac{2.56}{\sqrt{3.56}\sqrt{3.56}} = 0.719$$

> **Note**
>
> The item vectors are $\mathbf{s}_{i_1}$ and $\mathbf{s}_{i_5}$; their entries are user-centred ratings for the shared raters. This is the same cosine operation, but across item columns rather than user rows.

From Item Neighbours to a Prediction

$$\hat{r}_{ut} = \frac{\sum_{i \in Q_t(u)} \mathrm{sim}(t,i)\, r_{ui}}{\sum_{i \in Q_t(u)} |\mathrm{sim}(t,i)|}$$

$Q_t(u)$ — valid item neighbours
of target t that user u rated

Similarity $\mathrm{sim}(t,i)$
adjusted-cosine similarity between target and neighbour item

Observed rating r_{ui}
the user's rating of that neighbour item

> **Note**
>
> The neighbourhood changes with the target item and with the ratings that the user has actually supplied.

Target: recommend i_5 to u_1

Valid evidence:

$$Q_{i_5}(u_1) = \{i_1, i_3, i_4\}$$

These are items that **both** resemble target item i_5 and have an observed rating from u_1.

Weight u_1's rating of each item by its similarity to i_5.

Similarity IV: Significance Weighting

$$\text{discounted-sim}(u, v) = \text{sim}(u, v) \cdot \frac{\min\{|I_u \cap I_v|, \beta\}}{\beta}$$

Base Similarity: Pearson, cosine, or another chosen score.

Evidence Count: how many items both users rated.

Threshold (β): the overlap at which no further discount applies.

> 🤔 **To think about**
>
> This is a discount factor for users. Does it make sense for items?

Interpretation

One striking shared rating should not carry the same weight as twenty.

Role

This is a confidence modifier, not a replacement for the underlying similarity measure.

Worked Example IV: Significance Weighting

User–user example: users 1 and 6

Step 1 — start from the user-user Pearson score

$$\text{Pearson}(u_1, u_6) = -0.189$$

Step 2 — count the overlap and set a threshold

$$|I_{u_1} \cap I_{u_6}| = 3 \qquad \beta = 10$$

Step 3 — apply the confidence discount

$$\text{discounted-sim}_{\beta=10} = -0.189 \left(\frac{3}{10} \right) = -0.057$$

User–User KNN in RecBole

```python
from recbole.quick_start import run_recbole

run_recbole(
    model="ItemKNN",
    dataset="ml-100k",
    config_dict={
        "knn_method": "user",
        "k": 50,
        "shrink": 100,
    },
)
```

After training: score user 196 on item 242

```python
user = dataset.token2id["user_id"]["196"]
item = dataset.token2id["item_id"]["242"]
query = Interaction({
    "user_id": torch.tensor([user]),
    "item_id": torch.tensor([item]),
}).to(config["device"])
score = model.predict(query)
```

> **Note**
>
> `196` and `242` are MovieLens 100K raw IDs. Convert them to RecBole's internal IDs before prediction.

`knn_method: user`
Compare user rows, instead of its default items.

`k: 50`
Retain 50 neighbours.

`shrink: 100`
Regularise thin overlap.

`model.predict(query)` returns the estimated score for that one user–item pair. A recommender would repeat this over candidate items, then rank them.



👥 Discussion: Which Similarity Axis?

When would user-based or item-based similarity be more effective?

User-based: compare rows

When might a person's nearest neighbours be reliable enough to recommend from?

Item-based: compare columns

When might relationships between items be more stable or useful?

03:00

▶ Start Timer

Pairs: choose a setting, make a claim for one axis, then give one condition that could reverse your choice.
Optional: Consider User Archetypes ♣ · ♥ · ♦ · ♠

🔑 Key Point: Similarity Is a Modelling Choice

Choose similarity for the evidence you have—not because one metric is universally best.

Pearson

Corrects for rating generosity; thin overlap is still unstable.

Raw cosine

Keeps raw magnitude; it also keeps scale bias.

Adjusted cosine + weighting

Changes the comparison axis or discounts weak overlap.

🤞 **Ask which entries, baselines, and confidence assumptions shape the neighbours.**

> 🤔 **To think about**
>
> We can use similarity thresholds to calculate a neighbourhood instead of truncating to top k. When is that a good idea?

Neighbourhood Choices

Who enters the neighbourhood?

Top-k
Keep the closest fixed number of valid neighbours.

Similarity threshold
Keep everyone above a meaningful score; the neighbourhood size can vary.

Filter weak or negative scores
Avoid letting low-confidence or oppositional evidence dominate an estimate.

How should their evidence count?

Mean-centred prediction
Account for different user rating baselines.

Z-score normalisation *(not covered)*
Also account for how widely each user uses the rating scale.

Other variants
Amplify stronger similarities or aggregate votes for categorical ratings.

> 🤔 **To think about**
>
> Which choice changes *who* is eligible, and which changes *how* their evidence is aggregated?

From Neighbours to a Prediction

$$\hat{r}_{uj} = \mu_u + \frac{\sum_{v \in P_u(j)} \mathrm{sim}(u, v)\left(r_{vj} - \mu_v\right)}{\sum_{v \in P_u(j)} |\mathrm{sim}(u, v)|}$$

Neighbourhood $P_u(j)$: the top-k similar users who rated item j.

Similarity weight: closer neighbours contribute more.

Baseline and deviation: predict a deviation, then add the target user's own mean back.

Pipeline

1. choose a similarity
2. select valid top-k neighbours
3. aggregate their evidence
4. rank candidate items by $\hat{r}_{uj}$



Practicality: Making Neighbourhoods Work

Offline: prepare the neighbourhoods

- Compute user-user or item-item similarities.
- Store selected peer groups rather than every possible comparison when appropriate.
- Refresh them as ratings, users, and items change.

> **Note**
>
> User-based similarity scales with user pairs; item-based similarity scales with item pairs.

Online: serve a recommendation

- Combine the target's valid neighbours to make one prediction.
- Score many candidate items to produce a ranked top-k list.
- Balance latency, memory, freshness, and coverage—not accuracy alone.

> **To think about**
>
> Which work can be done before a request arrives, and which work must wait for the target user and item?

Section 05 Summary: Neighbourhood CF

1. Represent
Start with sparse user-item evidence and make missingness visible.

2. Compare
Choose an axis, similarity, centring rule, and confidence treatment.

3. Predict & rank
Select valid neighbours, aggregate evidence, then rank candidates.

Neighbourhood CF is transparent and local—but its outputs inherit the data's sparsity, exposure patterns, and modelling choices.



🔑 Summary

A recommender is never only an algorithm: it is a choice about evidence, objectives, and consequences.

🔑 Three ideas to carry forward

- **Frame:** define users, items, evidence, and usefulness.
- **Compare:** neighbourhood CF turns sparse overlap into a local estimate.
- **Question:** ask who is missing and whose outcome is served.



NEXT WEEK PRE-LECTURE EXERCISE

Look for Hidden Structure

Canvas discussion board · post before Mon, 23:59 SGT

Post a short, concrete response to this question; then reply constructively to one classmate.

> **Name one latent dimension that could explain several ratings—but is not an item metadata label.**

Next week: turn this intuition into latent-factor representations of the user-item matrix.



NEXT WEEK PREVIEW — WEEK 02

Latent Factor Models

What if useful similarities are not directly visible in a sparse matrix?

- **Move beyond local overlap** — infer compact user and item representations from many ratings together.
- **Interpret sparse evidence** — ask which hidden preferences may explain observed patterns.
- **Keep the same questions** — what evidence is represented, what does the model optimise, and who might it leave unseen?

